

Digital Signal Processing

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1 Introduction

Digital Signal Processing (DSP) is widely used in modern engineering applications such as telecommunications, audio processing, biomedical systems, and multimedia technologies. This laboratory work aims to study and implement fundamental DSP techniques using MATLAB.

The project focuses on signal analysis in both time and frequency domains, digital filtering methods, and the processing of noisy audio signals. MATLAB tools such as FFT analysis, Butterworth filtering, frequency response visualization, and filter design functions are used throughout the work.

The objective is to strengthen the understanding of DSP theory through practical implementation and numerical simulations.

2 Practical Work

This section applies the main concepts of digital signal processing using MATLAB. It is divided into two parts: sinusoidal signal filtering and audio signal processing.

The work includes frequency analysis, digital filter design, and performance evaluation using Fourier transform methods and MATLAB filtering functions.

Part - A Signal Filtering

A signal composed of multiple sinusoidal components with different frequencies is analyzed in both time and frequency domains. The objective is to remove a specific frequency component using an appropriate digital filter.

This part highlights the relationship between a signal and its frequency spectrum, as well as the effect of filtering on signal components. MATLAB functions such as `fft`, `fftshift`, `freqz`, `butter`, and `filter` are used for implementation and analysis.

A.1 Signal Generation, Spectrum Analysis and Filtering

A composite sinusoidal signal is generated with frequencies $f_1 = 5$ Hz, $f_2 = 20$ Hz, and $f_3 = 35$ Hz, sampled at $F_e = 512$ Hz over $t \in [0, 1]$ s.

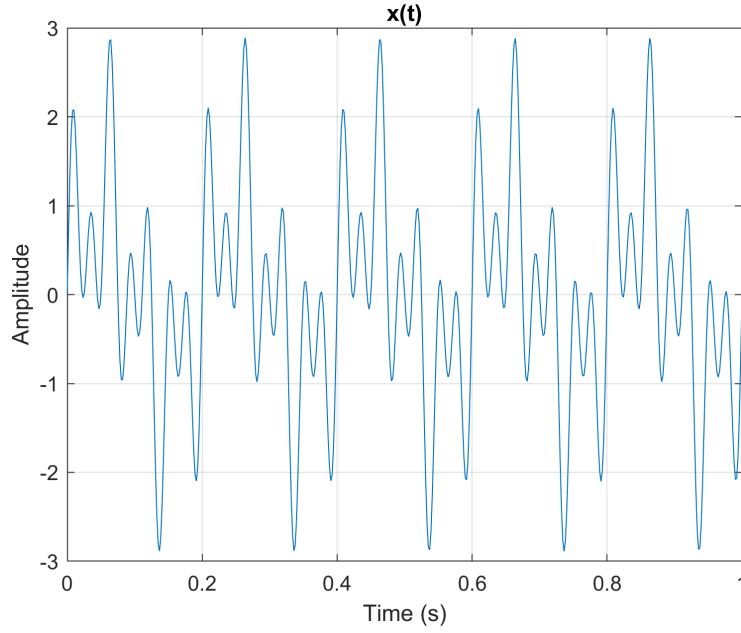


Figure 1: Time-domain signal $x(t)$

The frequency content of the signal is analyzed using the Fast Fourier Transform (FFT), showing the three main frequency components.

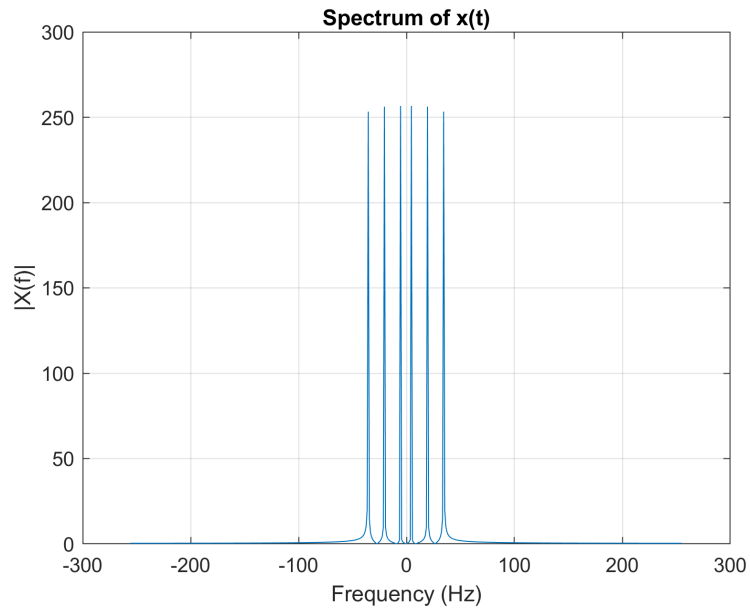


Figure 2: Spectrum of the original signal

A Butterworth band-stop filter is designed to remove the 20 Hz component while preserving the remaining frequencies. The Butterworth filter is chosen because it provides a maximally flat frequency response in the passband, avoiding ripples and preserving the original signal quality. It also offers good stability and efficient attenuation of unwanted frequency components, making it well suited for signal filtering applications.

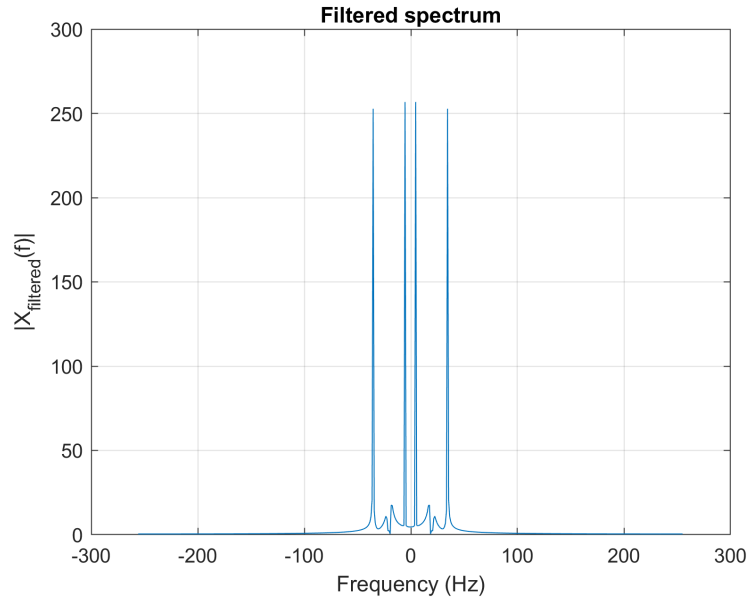


Figure 3: Spectrum after band-stop filtering

The results confirm that the 20 Hz component has been successfully attenuated.

A.2 Audio Signal Processing

This part focuses on the denoising of corrupted audio signals using digital filtering techniques. Two signals are analyzed: *bonjour_bruite.wav* and *mozart_bruite.wav*. The objective is to remove narrowband noise components identified in the frequency domain.

Bonjour signal analysis

The noisy signal is first analyzed in the time domain.

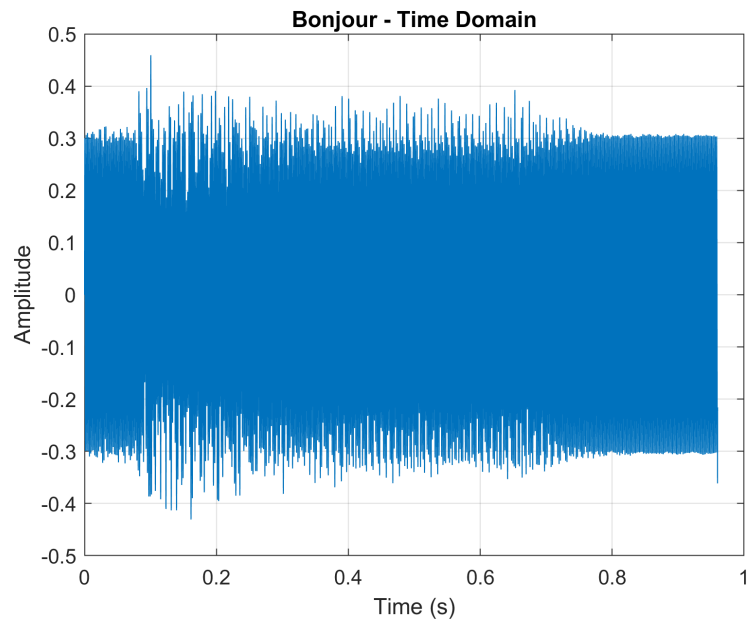


Figure 4: Noisy Bonjour signal in time domain

The frequency spectrum reveals a strong narrow peak around 500 Hz, indicating a tonal interference.

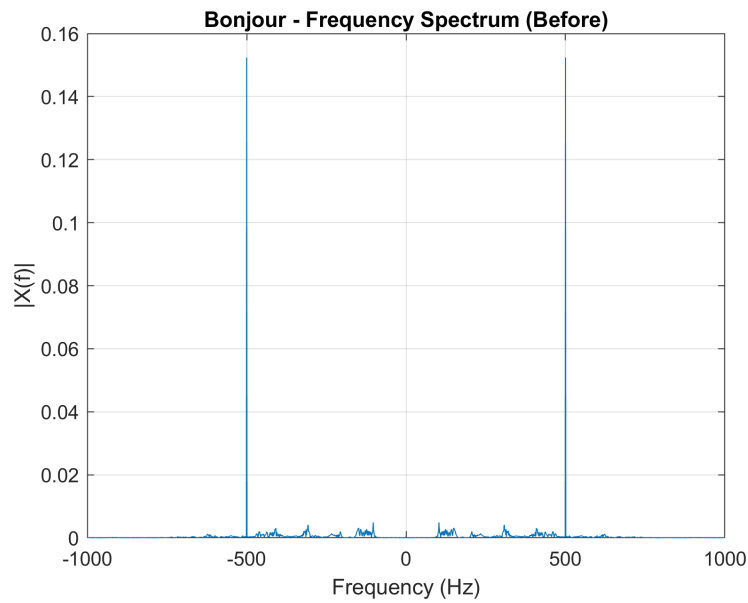


Figure 5: Bonjour spectrum before filtering

A Butterworth band-stop filter is designed to remove the 495–505 Hz component.

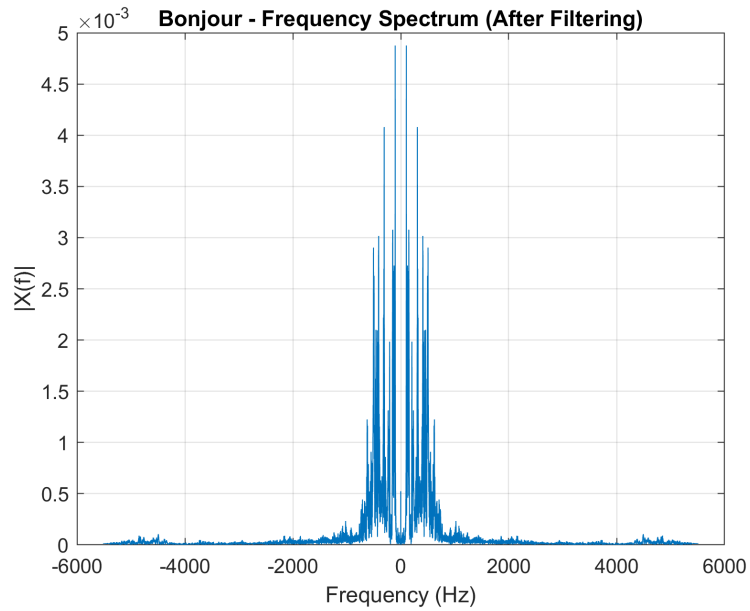


Figure 6: Bonjour spectrum after filtering

The filtered signal successfully removes the dominant interference while preserving the speech content.

Mozart signal analysis

The Mozart signal is then analyzed in a similar way.

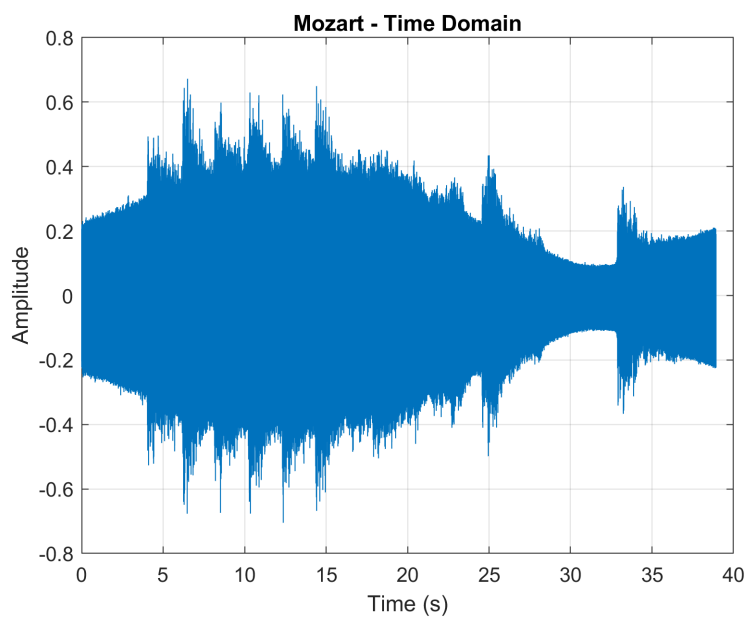


Figure 7: Noisy Mozart signal in time domain

The spectrum shows a strong tonal component around 700 Hz and additional high-frequency interference.

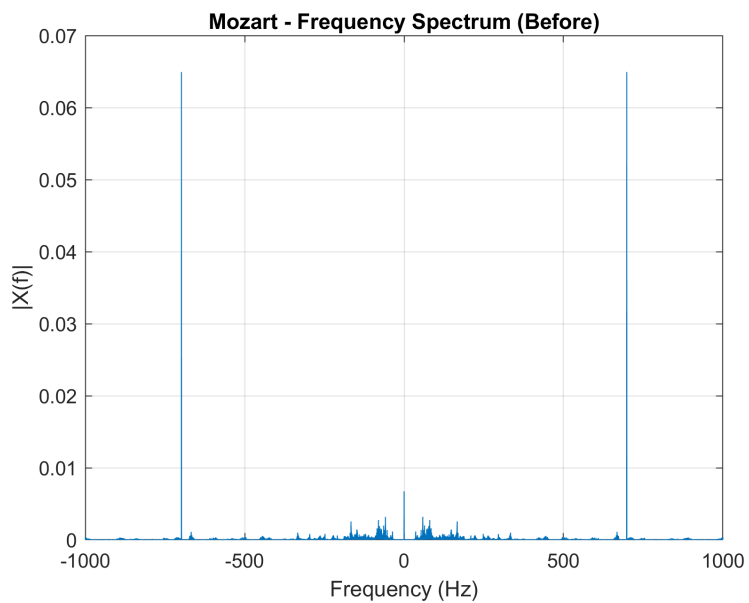


Figure 8: Mozart spectrum before filtering

A first band-stop filter is applied around 680–710 Hz to remove the dominant peak.

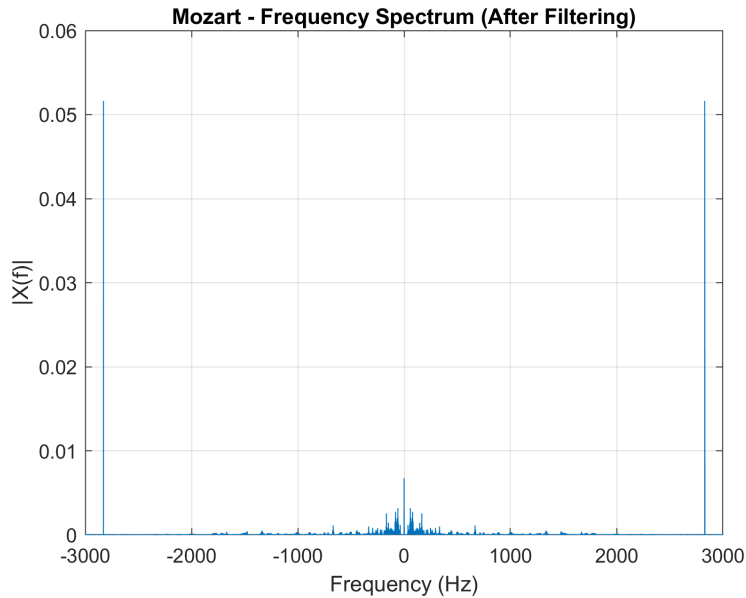


Figure 9: Mozart spectrum after filtering

After filtering, a residual interference peak is still observed around 2830 Hz, which is removed through a second filtering stage.

Conclusion

Digital filtering significantly improves signal quality by removing narrowband noise components while preserving the useful audio information. Frequency-domain analysis is essential for identifying unwanted components before filter design.

Part - B Digital Signal Processing

This section focuses on discrete-time signal processing and digital filter analysis using MATLAB. The objective is to study the representation, transformation, and filtering of discrete signals.

The exercises cover basic signal construction using elementary sequences such as impulse and step functions, followed by the analysis of digital filters described by difference equations or transfer functions.

Key tools such as `stem`, `zeros`, `ones`, `impz`, `freqz`, `filter`, `tf`, and `bodeplot` are used to visualize and analyze system behavior in both time and frequency domains.

B.1 Discrete-Time Signal Representation

This exercise focuses on the graphical representation of discrete-time signals using MATLAB. The objective is to illustrate basic operations such as impulse generation and rectangular signals using fundamental functions like `zeros`, `ones`, and `stem`.

Two discrete signals are considered:

$$x_1[n] = 0.3\delta[n-2] - 2\delta[n+1]$$

and a rectangular signal defined as:

$$x_2[n] = 1 \quad \text{for } 0 \leq n \leq 10$$

Signal representation

The signals are implemented using discrete index manipulation in MATLAB and plotted using the `stem` function.

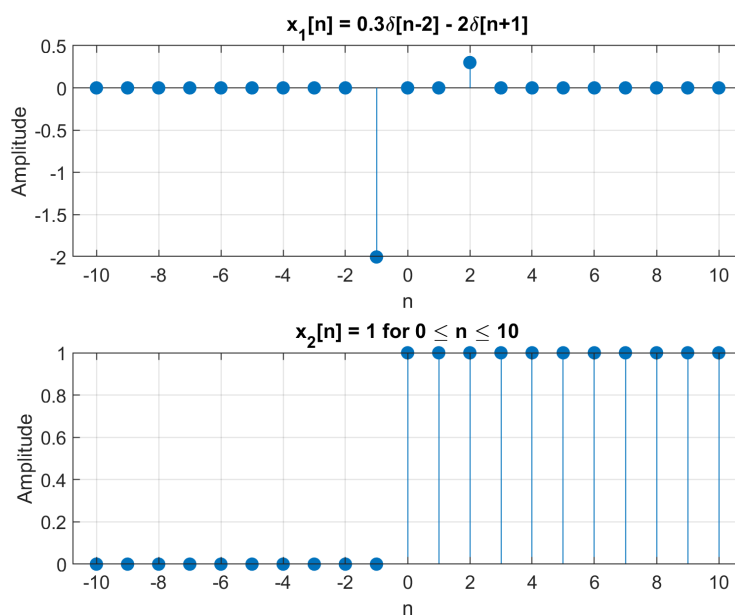


Figure 10: Discrete-time signals representation using subplot

The first signal consists of two shifted impulse functions, while the second signal represents a finite rectangular sequence.

Conclusion

This exercise illustrates how elementary discrete signals can be constructed and visualized using simple MATLAB operations. These basic building blocks are essential for understanding more complex digital signal processing systems.

B.2 Digital Filter Analysis

Now we analyze a discrete-time system defined by:

$$y[n] = 0.5 y[n - 1] + x[n] \quad (1)$$

The system is implemented as a first-order IIR filter and studied in time, frequency, and Z domains.

Impulse Response

The impulse response is computed using `impz` and shows an exponential decay, confirming a stable system.

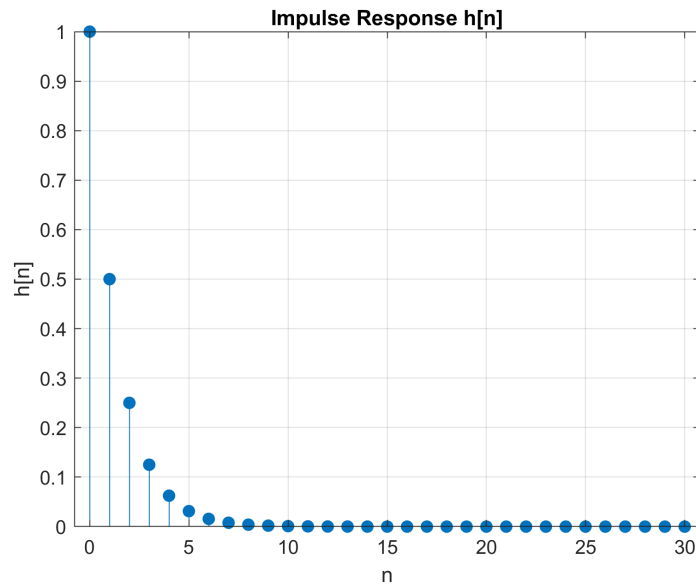


Figure 11: Impulse response $h[n]$

Transfer Function

The system is represented in the Z-domain as:

$$H(z) = \frac{1}{1 - 0.5z^{-1}} \quad (2)$$

Pole-Zero Analysis

The pole-zero map shows a single pole at $z = 0.5$, located inside the unit circle, which guarantees stability.

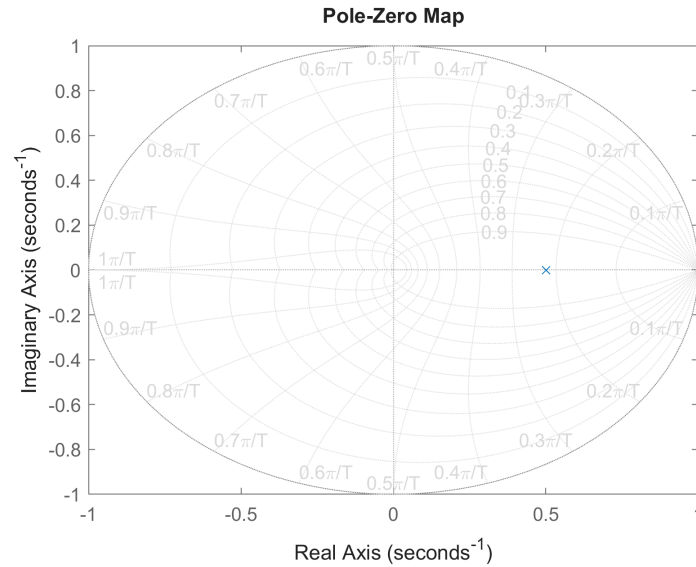


Figure 12: Pole-zero map of the system

Frequency Response

The frequency response computed using `freqz` shows a low-pass behavior, where low frequencies are preserved and high frequencies are attenuated.

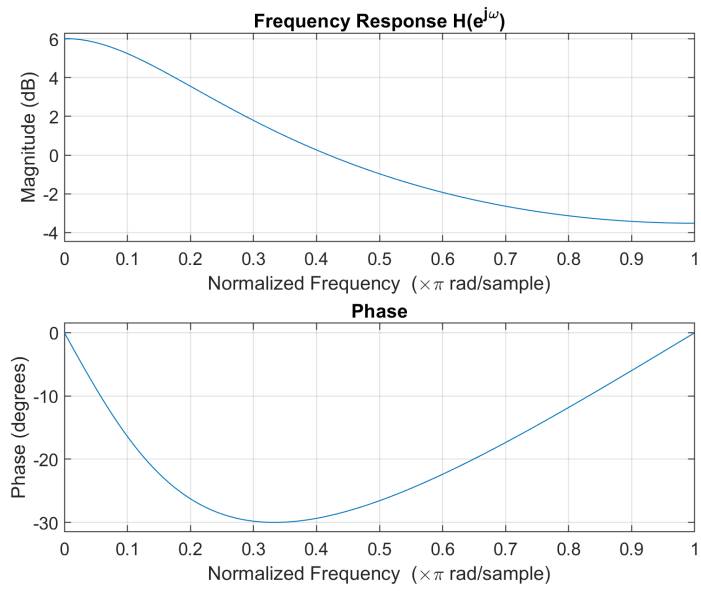


Figure 13: Frequency response $H(e^{j\omega})$

System Response

The system is tested with a step and sinusoidal input. The output is smoother due to the filtering effect.

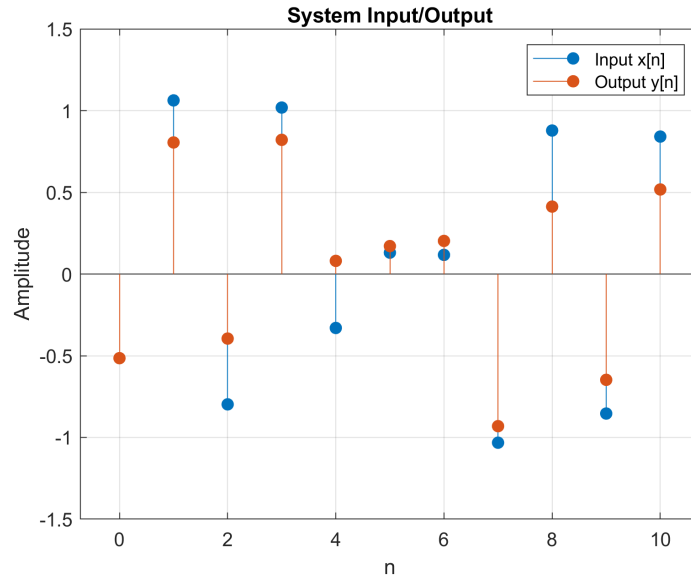


Figure 14: Input vs output signal

Conclusion

The system is a stable, causal first-order IIR low-pass filter that attenuates high-frequency components while preserving low-frequency content.

B.3 Digital Filter Analysis

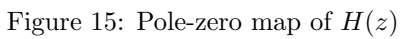
We consider the discrete-time system defined by the transfer function:

$$H(z) = \frac{z - 1}{z^2 + 20z + 75}$$

The objective is to analyze the system in the Z-domain, time domain, and frequency domain using MATLAB.

Pole-Zero Map

The pole-zero representation is obtained using `pzmap`. The poles are computed using `roots`.



Impulse Response

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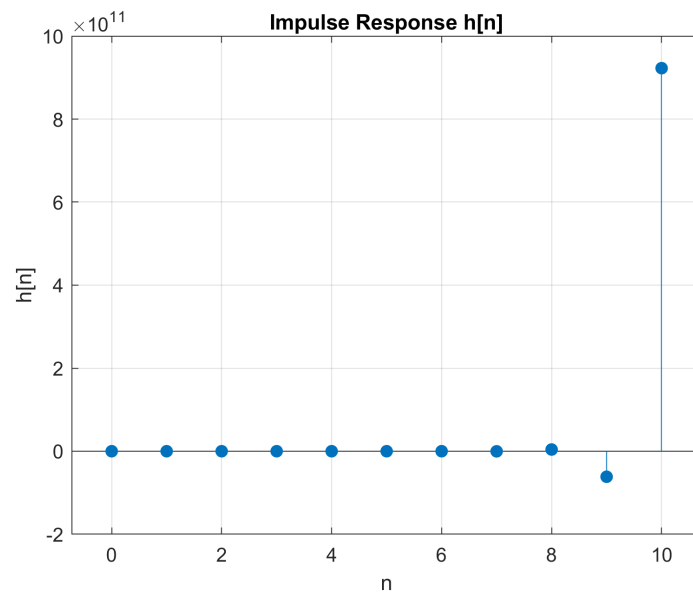


Figure 16: Impulse response $h[n]$

Frequency Response

The frequency response is obtained using `freqz`, showing the behavior of the system in the frequency domain.

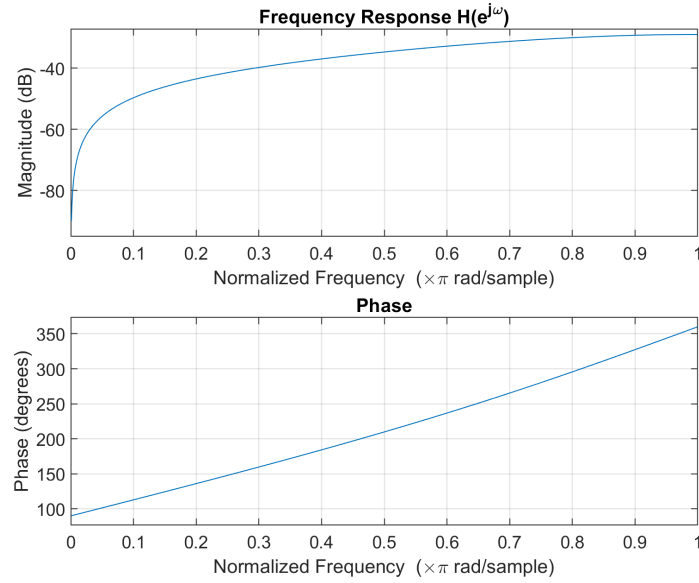


Figure 17: Frequency response $H(e^{j\omega})$

d) Stability Analysis

The poles of the system are:

$$z_1 = -5, \quad z_2 = -15$$

Since both poles lie outside the unit circle, the system is unstable.

Conclusion

The system is causal but unstable. Its impulse response is unbounded due to the location of poles outside the unit circle. Therefore, it cannot be used as a stable digital filter in practical applications.

General Conclusion

This laboratory work provided a comprehensive introduction to Digital Signal Processing (DSP) through practical MATLAB implementations. Time-domain and frequency-domain analyses were used to study both continuous audio signals and discrete-time systems.

The results demonstrated the effectiveness of Fourier analysis in identifying unwanted frequency components and the importance of digital filters in signal enhancement. Butterworth filters were successfully applied to remove narrow-band noise while preserving the useful signal content.

In addition, discrete-time system analysis using transfer functions, pole-zero representations, and frequency responses highlighted key concepts such as stability, causality, and system behavior in the Z -domain.

Overall, this work strengthened the understanding of theoretical DSP concepts through numerical simulation and practical signal processing applications.